AARAV SHARMA

Electronics & Communication Engineering Undergraduate

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EDUCATION

National Institute of Technology (NIT) Expected Graduation: May 2027

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering

- **Current CGPA:** 8.5 / 10.0
- **Relevant Coursework:** Microcontrollers & Embedded Systems, Digital Electronics, VLSI Design, Signal Processing, Computer Networks, Data Structures & Algorithms.

PROFESSIONAL EXPERIENCE

Embedded Systems Intern May 2025 – July 2025

Nexus Electronics Solutions Pvt. Ltd.

- Developed firmware using Embedded C for an ESP32-based automated environment monitoring module equipped with I2C-based digital sensor interfaces.
- Programmed standard peripheral protocols (UART, SPI, I2C) to establish zero-error hardware data packet collection.
- Optimized sensor polling subroutines and power-saving sleep cycles, decreasing average device power intake by 12%.
- Collaborated across functional hardware testing teams to prototype circuits on breadboards and troubleshoot signal continuity issues.

TECHNICAL PROJECTS

Smart IoT Asset Tracking System Winter 2025

Academic Hardware Project

- Engineered a tracking node utilizing an Arduino board integrated with GPS and GSM communication modems.
- Wrote software configurations using Python to parse NMEA location string parameters and automatically post geo-coordinates via custom HTTP requests.
- Designed safe power delivery pathways and enclosed the prototype chassis for secure industrial field environment deployment.

8-Bit ALU Circuit Simulation October 2025

Lab Project

- Modeled an 8-bit Arithmetic Logic Unit (ALU) supporting 16 logical operations using Verilog HDL.
- Validated operations, timing delays, and output registers across testbenches using ModelSim simulation suites.

TECHNICAL SKILLS

Hardware Systems:	8051 Microcontroller, Arduino, ESP32 Microcontrollers, FPGA Basics
Programming & Languages:	Embedded C, C++, Python, Verilog HDL, SQL Foundations
Software Tools:	MATLAB, Proteus Simulation, KiCAD PCB, ModelSim, Git

EXTRACURRICULAR ACTIVITIES & ACHIEVEMENTS

- **Technical Coordinator:** Managed core hardware lab logistics and micro-controller setups for 150+ students during the annual departmental engineering fest.
- **Participant:** Secured 3rd place in an intra-college circuit debugging competition evaluating rapid multi-meter validation and schematic correction skills.